Lecture 4: Foundations: Skinner

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-4 (CC BY-NC-SA 3.0).

Before you start

The lecture starts with an idea about why an unconscious mind might have evolved. Most of it is about behaviorism, the theory made famous by B. F. Skinner, and its three kinds of learning: habituation, classical conditioning and operant conditioning. At the end, Professor Bloom explains why almost everyone now agrees that the main claims of behaviorism are mistaken.

Learning goals

After this lecture you should be able to:

- Explain the deception theory of why an unconscious might have evolved.
- Describe the three main claims of behaviorism.
- Explain habituation, classical conditioning and operant conditioning, with an example of each.
- Identify the unconditioned and conditioned stimulus and response in an example.
- Describe the main evidence against behaviorism, including Chomsky's criticism.

Key ideas

1. A Brief Review on the Unconscious

- Most of what our minds do is unconscious. So a better question may be why a small part of mental life is conscious.
- One evolutionary answer is **deception**. Animals deceive: a threatened chimpanzee's hair stands up so it looks bigger. Humans are "masters of deception," but we are also good at detecting lies.
- "The best lies are lies we tell ourselves." If you believe your own lie, you will not give it away. Alfred Hitchcock made a child actor cry for real by whispering that his parents had left him.

2. B. F. Skinner and Behaviorism

- Behaviorism existed before Skinner (for example, John Watson). Skinner developed and publicized it, and he was very famous in the United States in the 1960s and 1970s.
- Behaviorism has three radical claims:
  1. **Learning is everything.** There is no real human nature. Watson said: "Give me a dozen healthy infants..." and he could train any of them to be any type of specialist.
  2. **Anti-mentalism.** Talk about desires, goals and emotions is unscientific. Science should use observable things: stimulus, response, reinforcement and punishment.
  3. **No interesting differences across species.** So we can study human learning by studying animals like rats and pigeons.

3. Habituation: The Very Simplest Form of Learning

- **Habituation** is "a decline in the tendency to respond to stimuli that are familiar due to repeated exposure." A sudden shout startles you, but less the second time. We stop noticing a ticking clock.
- It is useful: we should notice new things, which might be dangerous, but stop noticing things that are a normal part of the environment.
- Psychologists use habituation to study creatures who cannot talk, like animals and young babies.

4. Classical Conditioning: Associating Stimulus

- **Classical conditioning** is learning an association between two stimuli (events in the environment, like a sound or smell). Pavlov discovered it by chance while studying saliva in dogs.
- Food in the mouth (unconditioned stimulus, **UCS**) causes saliva (unconditioned response, **UCR**). A bell is at first neutral. After the bell and food are paired many times, the bell alone (conditioned stimulus, **CS**) causes saliva (conditioned response, **CR**).
- If the bell keeps coming with no food, the response goes away (**extinction**). After a wait, it can come back (**spontaneous recovery**).
- **Little Albert:** John Watson made a loud bang while a baby played with a rat. The baby became afraid of the rat, and the fear extended to a white rabbit (**stimulus generalization**).
- Uses: a theory of phobias (for example, after a dog bite), **systematic desensitization** (meeting the feared thing while relaxed), and cues that make people hungry.
- The modern view is that conditioning is **preparation**: the CS signals that the UCS is coming. A child who is beaten flinches when his father raises his hand. In *Clockwork Orange*, a drug (UCS) causes nausea (UCR); later, violence (CS) causes nausea (CR).

5. Operant Conditioning: Operating on the Environment

- **Operant** (or instrumental) **conditioning** is learning which of your actions work. Unlike classical conditioning, it involves voluntary actions.
- Thorndike's cats in a puzzle box did not solve it by insight. They tried many behaviors and slowly kept the one that worked. Skinner compared this to natural selection. The **law of effect**: an action becomes more likely if rewarded, weaker if not.
- **Negative reinforcement** is a kind of reward, not punishment: you take away something unpleasant, like removing a pig's heavy collar.
- **Shaping** means rewarding steps closer and closer to a complex behavior, such as teaching a pig to dance. Skinner taught animals to play Ping-Pong.
- "Good dog" becomes rewarding through classical conditioning. **Token economies** use rewards like poker chips.
- Rewards can follow **schedules**: fixed or variable, and ratio (number of actions) or interval (time). A slot machine is variable ratio. The **partial reinforcement effect**: behavior that is rewarded only sometimes lasts longer. This explains nagging.

6. Question and Answer on Behaviorism

A student noted that rewards like poker chips should lose their power over time (extinction), and Professor Bloom agreed. He added that some mental institutions and prisons have used token economies.

7. Controversies and Criticisms on Behaviorism

- Today "just about everybody agrees" that all three claims of behaviorism are mistaken.
- **Not everything is learned:** there is much evidence for innate knowledge and desires.
- **Mental states are scientific:** physics and chemistry are full of things you cannot see.
- **Reward is not necessary:** in Tolman's study, rats learned a maze even without reward. Rats seem to have **mental maps**.
- **Species have natural responses:** it is easy to train a pigeon to peck for food, but hard to train it to peck to escape. The **Garcia effect**: an animal avoids a new food if it then feels sick, but not if it then gets a painful shock. Taste and nausea are naturally linked.
- **Phobias:** humans and chimpanzees easily develop a fear of snakes. Phobias reflect our evolutionary history more than personal history.
- **Chomsky's "Review of Verbal Behavior":** when Skinner applies "reinforcement" to humans, the words become so vague that they are not science.
- Legacy: these learning mechanisms are real, and behaviorist methods help train animals and teach young children and people with severe autism.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
behaviorism	A theory that explains behavior through learning, without mental states	Skinner and John Watson
habituation	Responding less to something after it happens again and again	Getting used to a ticking clock
classical conditioning	Learning that one stimulus goes with another	The bell comes to cause saliva
unconditioned stimulus (UCS)	Something that causes a response without learning	Food in the dog's mouth
conditioned stimulus (CS)	A neutral thing that causes a response after learning	The rat for Little Albert
extinction	A learned response disappearing when it is no longer reinforced	The bell with no food
operant conditioning	Learning which actions bring good or bad results	Thorndike's cats in a puzzle box
law of effect	Rewarded actions grow stronger; unrewarded ones grow weaker	The cat keeps pressing the lever

Term	Meaning in simple English	Example from the lecture
negative reinforcement	Rewarding by taking away something unpleasant	Removing the pig's heavy collar
shaping	Rewarding small steps toward a complex behavior	Teaching a pig to dance
partial reinforcement effect	Behavior lasts longer when it is rewarded only sometimes	A slot machine; nagging
Garcia effect	Learning to avoid a food that came before sickness	Professor Bloom and the smell of ouzo

Check your understanding

1. Why might it help us to keep some motives unconscious?
2. What are the three main claims of behaviorism?
3. Why is habituation useful?
4. In Pavlov's experiment, what are the UCS, UCR, CS and CR?
5. In the Little Albert study, what are the CS and the CR? What is stimulus generalization?
6. What is the difference between negative reinforcement and punishment?
7. A parent says "no" to a child many times but sometimes says "okay." Why does the child keep asking?
8. Explain the Garcia effect and why it is a problem for behaviorism.

Answer guide

1. The best lies are lies we tell ourselves. If we do not know our own sinister motives, we will not give them away to others.
2. Everything is learned; mental states are unscientific; and learning works the same way in all species.
3. We notice new things, which might be dangerous, but stop noticing things that are a normal part of the environment.
4. UCS: food in the mouth. UCR: saliva. CS: the bell. CR: saliva caused by the bell.
5. The CS is the rat and the CR is fear. Stimulus generalization means the fear spread to similar things, like a white rabbit.
6. Negative reinforcement is a type of reward: you take away something unpleasant, like the pig's heavy collar, when it does what you want. Punishment is for behavior you do not want, like the pig walking backward.
7. This is the partial reinforcement effect. Behavior that is rewarded only sometimes lasts longer.
8. Animals link a new taste with sickness, but not with a painful shock. So not all stimuli and responses are equal; animals have natural, built-in tendencies.

English practice

Useful phrases from the lecture

- "At the core of... are..." - names the most important parts of a theory. Example: "At the core of behaviorism are three extremely radical and interesting views."
- "It turns out that..." - introduces a surprising finding. Example: "It turns out that there are certain phobias that we're specially evolved to have."
- "This is actually no longer the mainstream view." - says that most experts have changed their opinion.
- "I think it's fair to say that..." - gives a careful but confident judgment.

Speaking task (with your AI tutor)

- Explain classical conditioning in 4-6 sentences, using Pavlov's dog. Use these terms: *unconditioned stimulus*, *conditioned stimulus*, *extinction*.
- Explain why behaviorism lost its power in 4-6 sentences. Use these terms: *innate*, *mental maps*, *Garcia effect*.

Writing task

- Write one paragraph (80-120 words): Compare classical conditioning and operant conditioning. Say what is learned in each, whether the learner must act, and give one example of each from the lecture.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 4. Ask me questions about classical and operant conditioning, one at a time."
- "Quiz me on the key vocabulary of Lecture 4."
- "Give me an example of conditioning, and I will name the UCS, UCR, CS and CR."
- "I will explain the partial reinforcement effect in my own words. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), chapter 4
- Chomsky, Noam. "A Review of B. F. Skinner's Verbal Behavior." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 129-137
- Watson, John B. and Rosalie Rayner. "Conditional Emotional Reactions." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 117-129

My notes